**Thm.** Let $K$ be a field, and $K_1, K_2 \le K$ be subfields.
Define a Composite field of $K_1$ and $K_2$ as:

$$K_1K_2 \overset{\text{def}}{=} \bigcap_{\substack{F \le K \\ K_1, K_2 \le K}} F$$

That is, the smallest subfield of $K$ containing $K_1 \cup K_2$.

**Lemma.** Let $K/F$ be a extension field, and let $\alpha_1, \alpha_2 \in K$. Then,

$$F(\alpha_1)F(\alpha_2) = F(\alpha_1, \alpha_2)$$

Proof. Since $F(\alpha_1, \alpha_2)$ contains both $F(\alpha_1)$ and $F(\alpha_2)$, $\subseteq$, and since $\alpha_1, \alpha_2 \in F(\alpha_1)F(\alpha_2)$, so proved.

**Thm.** Let $K/F$ be an Extension, and $K_1, K_2 \le K$ be finite extensions for $F$. Then,

$$[K_1K_2 : F] \le [K_1 : F] \cdot [K_2 : F]$$

Moreover, if $[K_1 : F], [K_2 : F]$ are relatively prime, then the equality hold.
Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be a $F$-basis for $K_1$, and $\{\beta_1, \cdots, \beta_m\}$ be a $F$-basis for $K_2$.
Then, clearly,

$$K_1K_2 = F(\alpha_1, \cdots, \alpha_n, \beta_1, \cdots, \beta_m).$$
$$= K_1(\beta_1, \cdots, \beta_m)$$

That is, $K_1K_2$ is spanned by $\beta_1, \cdots, \beta_m$ over $K_1$, hence

$$[K_1K_2 : K_1] \le m = [K_2 : F].$$

Now, multiplying $[K_1 : F]$ to the both side, we obtain the first assertion.
And, take $[K_1 : F] = n$ and $[K_2 : F] = m$. Since

$$[K_1K_2 : F] = [K_1K_2 : K_i][K_i : F] \quad (i = 1, 2)$$

so $[K_1K_2 : F]$ is divided by $n$ and $m$, so $\operatorname{lcm}(n, m) \mid [K_1K_2 : F]$.
And, if $\gcd(n, m) = 1$, then $\operatorname{lcm}(n, m) = 1$, so $nm \le [K_1K_2 : F]$.

Example.

Let $K = \mathbb{Q}(\sqrt[4]{2}, \sqrt{3}, \sqrt{6})$ and $K_1 = \mathbb{Q}(\sqrt[4]{2}, \sqrt{3})$ and $K_2 = \mathbb{Q}(\sqrt{6})$.

Then, $K = K_1 K_2$. And, $[K_1 : \mathbb{Q}] = 8$, and $[K_2 : \mathbb{Q}] = 2$.

As we expect, $[K : \mathbb{Q}] = 8 \cdot 2 = 16$ ?

Since $K_1$ is $\mathrm{span}_{\mathbb{Q}}\{1, \sqrt[4]{2}, \sqrt{2}, \sqrt[4]{8}, \sqrt{3}, \sqrt[4]{2}\sqrt{3}, \sqrt{6}, \sqrt[4]{8}\sqrt{3}\}$,

and $K_2$ is $\mathrm{span}_{\mathbb{Q}}\{1, \sqrt{6}\}$,

So $K_2 \subseteq K_1$, $K_1 K_2$ has degree 8.

